

STATEMENT OF WORK

Body Plethysmograph – Rolla CBOC

1. Background

The Harry S Truman VA Medical Center, Multi-Specialty CBOC – Rolla, requires a Body Plethysmograph to outfit their new facility. This is a brand name or equal requirement.

2. Scope of Work

- a. Item: Body Plethysmograph (see technical specifications, Section 3)
- b. Brand Name or Equal Requirement (see Section 4)
- c. Installation (see Section 5)
- d. Training (see Section 6)
- e. Warranty and Support (see Section 7)

3. Technical Specifications

- a. Shall be a commercially available, non-developmental cardiopulmonary diagnostic testing system designed for pulmonary and cardiopulmonary function testing.
- b. Shall be suitable for use in hospital, clinic, or pulmonary laboratory environments.
- c. System shall support adult and pediatric patient populations.
- d. Shall be a modular system capable of supporting multiple pulmonary diagnostic tests within a single platform.
- e. Shall be designed for routine clinical testing, diagnostic evaluation, and disease management.
- f. The system shall support the following test modalities, either integrated or via modular expansion:
 - i. Spirometry (pre- and post-bronchodilator testing)
 - ii. Diffusion Capacity (DLCO) testing
 - iii. Lung Volumes using gas dilution and/or body plethysmography
 - iv. Cardiopulmonary Exercise Testing (CPET) capability
 - v. Metabolic gas analysis with breath-by-breath measurements
 - vi. Resting and exercise testing protocols
- g. Shall provide breath-by-breath gas exchange analysis.
- h. Shall measure oxygen uptake (VO₂), carbon dioxide production (VCO₂), and ventilation (VE).
- i. Shall support multiple gas analyzers with automated calibration.
- j. Shall include high-precision flow and volume measurement with real-time correction for temperature, pressure, and humidity.
- k. Shall support automated quality control and calibration routines.
- l. Shall allow custom and standardized exercise protocols.
- m. Shall provide real-time physiologic monitoring and display during exercise testing.

- n. Shall support ECG, blood pressure, and pulse oximetry integration (direct or interfaced).
- o. Shall include FDA-cleared diagnostic software for pulmonary and cardiopulmonary testing.
- p. Shall provide real-time graphical displays, automated calculations, and interpretation tools.
- q. Shall support user-configurable reports and standardized reference equations.
- r. Shall allow electronic storage, retrieval, and review of patient test data.
- s. Shall support network connectivity and integration with hospital information systems and electronic medical records (EMR).
- t. Shall include an intuitive graphical user interface suitable for clinical staff.
- u. Shall support guided workflows to ensure standardized testing and repeatability.
- v. Shall allow multi-user access with role-based permissions.
- w. Shall minimize setup and test time to support high patient throughput.
- x. Shall comply with applicable FDA regulations for diagnostic medical devices.
- y. Shall meet ATS/ERS standards for pulmonary and cardiopulmonary testing.
- z. Shall include infection control features, including compatibility with disposable patient interfaces.
- aa. Shall include system alerts and error checking to ensure data integrity and patient safety.
- bb. System shall include an enclosed body plethysmography chamber (body box) capable of measuring lung volumes and airway resistance in accordance with ATS/ERS standards.
- cc. Internal volume: 726 L
- dd. Construction: Aluminum/acrylic
- ee. Arm adjustments
 - i. Vertical extension: 20 in (51 cm)
 - ii. Horizontal extension: 14 in (35.5 cm)
- ff. Prevent flow sensor specs
 - i. Bidirectional Pitot tube flow sensor
 - ii. Range: ± 18 L/s
 - iii. Accuracy: $\pm 2\%$ or 50 mL, whichever is greater (meets or exceeds ATS/ERS clinical performance standards)
 - iv. Resistance: < 1.5 cm H₂O
 - v. 0/L/s @ 14 L/s
 - vi. Dead space: 39 mL
- gg. Power requirements: 100-240 V/50-60 Hz
- hh. O₂ analysis (model dx)
 - i. Type: Galvanic
 - ii. Range: 0-100%

- ii. Diffusion analysis: rtd multigas technology
 - i. Analysis time: <1 sec
 - ii. Range for CO and CH4: 0-0.35%
 - iii. Accuracy for CO and CH4: 0.003%
 - iv. Linearity: <1% full scale
 - v. Resolution for CO and CH4: $\pm 0.0005\%$
- jj. Gas sample
 - i. Proprietary gas-drying sample circuit
 - ii. GAS REQUIREMENTS
 - iii. Calibration gas (6-8 psi): 5% CO₂, 12% O₂, bal N₂
 - iv. DLCO mix (135 psi): 0.3% CO, 0.3% CH₄, 21% O₂, bal N₂
 - v. 100% O₂ (135 psi) - DX model only
- kk. All models require gas to seal the door
 - i. Air (30-135 psi) or DLCO mix (135 psi)

4. Brand Name or Equal Requirement

The requirement for this acquisition is based on the Platinum Elite DX Body Plethysmograph with RTD by MGC Diagnostics. The brand listed above is not specifically required, but serves as example to illustrate types of equipment that meet the required salient characteristics for this acquisition. Products considered “equal” must meet the salient physical, functional, or performance characteristics specified in this statement of work. The contractor is responsible for submitting sufficient documentation for the Government to evaluate “equal” items.

5. Installation

The contractor shall be responsible for the installation, performance verification, and validation of the structural components with a VA Activations member who will accept the equipment. Coordination will occur between COR, Activations PM, and the General Contractor for utility connections.

White glove installation shall be included for all products provided.

6. Training

The vendor shall coordinate through the VA POC to provide a hands-on training session to staff selected by the VA. Training shall be completed within 45 days after final installation of equipment. The vendor will record the training and provide an electronic copy of the training to the VA POC for VA to use to train additional personnel at a future date. In addition, the vendor shall provide additional training documents to the VA POC for training additional persons at a future date.

7. Warranty and Support

The contractor shall provide a manufacturer’s two-year, free replacement warranty on units with normal wear and tear, including free return shipping, from date of delivery.